



Multi-Model Electricity Load Forecasting for Delhi (2000–2023): A Comparative Analysis of Statistical, Ensemble, and Deep Learning Approaches

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Abstract

The increasing energy needs of Delhi, the capital of India, demand smart and responsive forecasting systems that can provide grid stability, optimal distribution, and forward-looking energy planning are few of the themes discussed. The given study provides an exhaustive comparative review of classical statistical models, machine learning methods, and deep learning methods applied to forecast electricity load forecasting using the hourly consumption data series ranging from 2000 to 2023. Various models used include ARIMA, SARIMA, XGBoost. The authors analyze the performance of Gradient Boosting, and LSTM networks in forecasting short-term electricity demand. This work puts great stress on feature engineering based on time-series characteristics and lag variables, and seasonality patterns to improve the accuracy of the forecast. RMSE, MAE, and MAPE measures in all the models. Outcomes indicate that though standard models perform well in stationary data, ensemble learning and deep neural networks show better performance when it comes to nonlinear consumption capture behavior. This effort is intended to help energy stakeholders to maximize load scheduling and to create real-time monitoring systems thereby enable data-driven smart grid policy in Indian cities.

Keywords: Electricity load forecasting, ARIMA, SARIMA, XGBoost, Gradient Boosting, LSTM, Delhi power consumption, Time series analysis, Feature engineering, Smart grid

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1. Introduction

Patterns of electricity consumption have a big impact on the power grid systems strategic management, operational effectiveness, and infrastructure maintenance. Prudent management based on intensity and consumption patterns is vital towards effectiveness and optimal operation of power systems [1].

Electricity use has various important socioeconomic effects. Economic development is largely driven by electricity, which powers commercial activity operations, industrial processes, and household needs. Electrical power runs many industrial manufacturing processes and machinery sectors, raising productivity and contributing to economic growth. Electricity is used in commercial establishments for processing transactions, running information technology, and providing necessary customer services [2]. Electricity powers lighting systems, climate control devices, communication devices, and household appliances. It forms the core basis of livelihood for residential consumers and is hence indispensable in everyday life.

Power Electricity Consumption (PEC) acts as a strong indicator of economic vitality. Industrialized nations rely heavily on abundant electrical supply for sustaining industries, domestic consumption, agricultural operations, and governmental functions [4]. Nations with limited domestic energy resources usually rely on international energy imports [5]. This situation demands proper planning and reliable

forecasts of electricity demand. However, electricity demand is difficult to predict, given that it has inherent volatility and the difficulty of direct physical measurement. Hence, robust models are crucial to ensure stability, in particular under uncertain or changing conditions [6].

Maintaining optimum energy supply-demand balance, reducing operational and maintenance costs, and preserving grid stability all depend on accurate PEC forecasting. Accurate demand forecasts are essential in future energy planning, system optimization, and the advancement of sustainable energy management since electricity cannot be stored efficiently [7].

1.1. Electricity Consumption in India: A Growing Demand

The fastest to grow, in terms of electricity consumption, has been India. This is attributed to urbanization, industrial growth, and rising household demands. The International Energy Agency forecasts that through 2040, demand for electricity in India will grow at a CAGR of 5%, or about twice the world average. Through the last two decades, consumption has steadily increased, yet it remains below that of the developed countries.

The sharp increase in demand underlines the urgent requirement for advanced electricity demand forecasting models specific to India's unique socioeconomic, climatic, and infrastructure features. The prediction problem is exacerbated by the high degree of regional variability, seasonal variations, and growing integration of renewable energy sources.

1.2. Research Objectives

The following are the main goals of this study:

1. Creation and comparison of several forecasting models for the patterns of electricity consumption in Delhi.
2. Analyzing the effects of feature engineering and lag variables on the model performance.
3. Evaluation of the efficacy of contemporary machine learning techniques versus conventional statistical methods.
4. Provision of actionable insights for energy policy makers and grid operators.

2. Literature Review

2.1. Forecasting Techniques: An Overview

Numerous models for predicting electricity consumption have been the subject of extensive research. These models can be broadly classified as statistical models (such as regression analysis, ARIMA, and Kalman filters), non-linear intelligent models (such as support vector machines and artificial neural networks), and grey prediction models (made for uncertain or incomplete data).

Improved performance has been shown by hybrid models that incorporate these strategies. For example, to increase accuracy in short-term forecasting applications, Bouzerdoum et al. [8] combined SARIMA and SVM. In Greece, Ekonomou [9] used ANN for long-term forecasting, outperforming conventional linear models. The usefulness of combining the ARIMA and Grey Models was also shown by Yuan et al. [11] and Ediger et al. [12].

Recent advancements include ensemble methods that combine statistical rigor with the adaptability of artificial intelligence, metaheuristic optimization algorithms, and deep learning techniques like Long Short-Term Memory (LSTM) networks.

2.2. Feature Engineering in Time Series Forecasting

Forecasting accuracy has significantly increased as a result of the addition of engineered features. By finding temporal dependencies that raw data might miss, research has shown that lag variables, seasonal decomposition, and cyclical patterns can improve model performance [13, 14].

2.3. Research Gap

Although there is much research about electricity forecasting, there is not a thorough comparison of statistical, machine learning, and deep learning techniques that particularly address Delhi's particular consumption patterns. The paper provides a systematic review across until now; very few studies have been performed by comparing several model categories. This study fills this gap.

3. A Historical View of India's Power Electricity Consumption

3.1. Early Development (1897–1947)

The introduction of electricity in Kolkata (formerly Calcutta) in 1897 marked the beginning of India's experience with electricity in the late 19th century. Electricity was initially considered a luxury rather than a necessity because it was restricted to urban areas and had few uses.

3.2. Post-Independence Expansion (1947–1980s)

India gave infrastructure development, including electrification, top priority after gaining independence in 1947. The foundation for a centralized electricity board system across states was laid by the Electricity (Supply) Act of 1948. Electric pumps and irrigation systems created demand in agriculture during the Green Revolution of the 1960s and 1970s.

3.3. Liberalization Era and Economic Growth (1990s–2000s)

Rapid industrialization and urbanization brought about by economic liberalization greatly increased the demand for electricity. Initiatives for privatization got underway, and the use of home appliances sharply increased.

3.4. 21st Century: Reforms and Renewable Push (2010s–Present)

Competitive market structures were made possible by the Electricity Act of 2003. The goal of major programs like Saubhagya and DDUGJY was to electrify the entire population. By 2030, India aims to reach 500 GW of non-fossil fuel capacity.

4. Methodology

4.1. Dataset Description

The Central Electricity Authority (CEA) and Delhi Transco Limited (DTL) provided the hourly electricity consumption data for Delhi from 2000 to 2023, which were used in this study. The dataset includes a time period of 24 years (2000–2023), with hourly granularity, approximately 210,240 total hourly observations, and variables consisting of timestamp, electricity consumption (MW), temperature, humidity, and seasonal indicators.

4.2. Data Preprocessing

Preprocessing of the data included: (1) Missing Value Treatment using linear interpolation for gaps shorter than six hours and forward-fill for longer periods; (2) Outlier Detection using the IQR method and domain knowledge to find unusual consumption patterns; (3) Stationarity Testing using the Augmented Dickey-Fuller test to confirm unit roots; and (4) Normalization using Min-Max scaling for neural network inputs.

113 4.3. Feature Engineering

114 The following features were created in order to record temporal patterns and dependencies: lag vari-
115 ables that show historical consumption levels at different time points ($t-1$, $t-2$, through $t-24$, $t-48$, and
116 $t-168$ for weekly lag); rolling statistics, such as means and standard deviations for the 24-hour, 7-day,
117 and 30-day rolling periods; weather variables including temperature, humidity, and derived cooling and
118 heating degree days; seasonal features including the hour of the day, day of the week, month, and season;
119 and cyclical features encoded using sine and cosine transformations.

120 4.4. Model Development

121 4.4.1. Statistical Models

122 ARIMA (AutoRegressive Integrated Moving Average): Seasonal decomposition analysis, residual di-
123 agnostics, and AIC and BIC criteria were used in order selection.

124 SARIMA (Seasonal ARIMA): Used grid search methodology to optimize parameters while incorpor-
125 ating seasonal patterns.

126 4.4.2. Machine Learning Models

127 XGBoost (Extreme Gradient Boosting): Cross-validation, feature importance analysis, and Random-
128 izedSearchCV were used to implement hyperparameter tuning for model selection.

129 Gradient Boosting Regressor: Early stopping and learning rate optimization were used in sequential
130 ensemble learning to avoid overfitting.

131 Random Forest: Used a decision tree ensemble with out-of-bag error estimation and bootstrap aggre-
132 gating.

133 4.4.3. Deep Learning Models

134 LSTM (Long Short-Term Memory): The architecture included a dense output layer, two LSTM layers
135 (128, 64 units), a look-back window of 168 hours (1 week), and dropout layers (0.2) for regularization.

136 GRU (Gated Recurrent Unit): Simplified architecture with bidirectional layers for improved context
137 and faster training with comparable performance when compared to LSTM.

138 CNN-LSTM Hybrid: Combined 1D Convolutional layers for local pattern extraction with LSTM layers
139 for sequential dependencies, creating a comprehensive feature learning architecture.

140 4.5. Experimental Setup

141 4.5.1. Train-Validation-Test Split

142 Data partitioning: Training set covering 2000–2019 (70% of data); Validation set spanning 2020–2021
143 (15% of data); Testing set encompassing 2022–2023 (15% of data).

144 4.5.2. Evaluation Metrics

145 Three primary metrics assessed model performance:

$$\text{RMSE} = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n}} \quad (4.1)$$

$$\text{MAE} = \frac{\sum |y_i - \hat{y}_i|}{n} \quad (4.2)$$

$$\text{MAPE} = \frac{100}{n} \times \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (4.3)$$

146 4.5.3. Cross-Validation Strategy

147 Time series cross-validation with expanding window, 5-fold validation maintaining temporal order,
148 and walk-forward validation for final model assessment were employed.

149 **5. Results and Analysis**

150 *5.1. Baseline Statistical Models Performance*

151 ARIMA Results: The ARIMA(2,1,2) model had moderate performance, with RMSE of 245.6 MW,
152 MAE of 189.3 MW, and MAPE of 12.4%. The model captured the fundamental trends, but encountered
153 difficulties with sudden demand spikes and seasonal variations.

154 SARIMA Results: SARIMA(2,1,2)(1,1,1)₂₄ exhibited a MAPE of 9.8%. Integration of seasonal com-
155 ponents significantly improved performance over plain ARIMA.

156 *5.2. Impact of Lag Variables and Feature Engineering*

157 The addition of lag variables and engineered features significantly improved all of the models. Ran-
158 dom Forest improved from 189.6 MW to 156.7 MW (17.3% improvement), Gradient Boosting increased
159 from 165.2 MW to 128.9 MW (22.0%), and XGBoost increased from 178.5 MW to 142.3 MW RMSE (20.3%
160 improvement).

161 Among the other lags, the 24-hr lag ($t-24$) provided the largest correlation at 0.89 with current con-
162 sumption, the weekly lag ($t-168$) captured weekly patterns, and rolling statistics reduced the impact of
163 noise by a smoothing effect.

164 *5.3. Machine Learning Models Performance*

165 XGBoost: XGBoost with engineered features reached strong performance, with a RMSE of 142.3 MW,
166 MAE of 108.7 MW, and MAPE of 6.9%. Feature importance analysis showed: 24-hour lag (23.4%), hour
167 of day (18.7%), 7-day rolling mean (15.2%), temperature (12.8%), and day of week (11.3%).

168 Gradient Boosting (Best Performing Model): Gradient Boosting with lag features had the best overall
169 performance: RMSE of 128.9 MW, MAE of 98.2 MW, and MAPE of 6.2%. The advantages included
170 sequential learning, capturing complex patterns, robustness to outliers and missing values, excellent
171 handling of nonlinear relationships, and strong performance across all seasons.

172 *5.4. Deep Learning Models Performance*

173 LSTM Networks: Standard LSTM architecture yielded RMSE of 156.4 MW, MAE of 119.6 MW, and
174 MAPE of 7.8%. The LSTM model effectively captured long-term dependencies but required extensive
175 hyperparameter tuning and proved computationally expensive.

176 GRU Networks: GRU demonstrated comparable performance with faster training: RMSE of 162.1
177 MW, MAE of 124.3 MW, and MAPE of 8.1%.

178 CNN-LSTM Hybrid: The combined architecture attempted to leverage both local and sequential pat-
179 terns: RMSE of 149.8 MW, MAE of 115.2 MW, and MAPE of 7.4%.

180 *5.5. Comparative Analysis*

Table 1: Model Performance Comparison.

Model	RMSE (MW)	MAE (MW)	MAPE (%)	Training Time
ARIMA	245.6	189.3	12.4	Low
SARIMA	198.7	156.8	9.8	Low
Random Forest	156.7	118.9	7.6	Medium
XGBoost	142.3	108.7	6.9	Medium
Gradient Boosting	128.9	98.2	6.2	Medium
LSTM	156.4	119.6	7.8	High
GRU	162.1	124.3	8.1	High
CNN-LSTM	149.8	115.2	7.4	Very High

181 5.6. Seasonal Performance Analysis

182 Different patterns were found when the model's performance was analyzed over several seasons.
183 Gradient Boosting achieved 6.8% MAPE during the summer (April–June), when consumption was at
184 its highest due to cooling demands, whereas traditional models had trouble with abrupt spikes. The
185 most erratic consumption patterns were seen during the monsoon season (July to September), when
186 all models showed higher error rates, although gradient boosting remained the most stable. The most
187 consistent consumption patterns were seen during the winter months of November through February,
188 when SARIMA demonstrated competitive performance and all models performed well.

189 6. Discussion

190 6.1. Key Findings

191 This investigation produced a number of important findings. All model performances were consid-
192 erably enhanced by 17–22% with the addition of lag variables, especially 24-hour and weekly lags. With
193 a 6.2% MAPE, gradient boosting with engineered features proved to be the best model, surpassing both
194 deep learning and conventional statistical techniques.

195 The limited dataset size in relation to model complexity, the need for extensive hyperparameter tun-
196 ing, and the higher computational overhead are likely the reasons why LSTM and GRU networks did
197 not outperform optimized ensemble methods, despite their theoretical advantages in capturing complex
198 patterns. Raw temporal features were not as valuable as engineered features, especially cyclical encodings
199 and rolling statistics.

200 6.2. Practical Implications

201 For day-ahead scheduling, real-time dispatch decisions, and peak load management, the 6.2% MAPE
202 achieved by gradient boosting offers satisfactory accuracy. Better unit commitment decisions, reduced
203 operational costs, and a reduction in the need for spinning reserves can together result from more ac-
204 curate forecasting. Demand-side management initiatives, time-of-use pricing plans, and infrastructure
205 investment choices can all be informed by insights derived from feature importance.

206 6.3. Model Interpretability vs. Performance Trade-off

207 The Gradient Boosting model offers the best balance between predictive accuracy, model interpretabil-
208 ity, computational efficiency, and implementation complexity, even though deep learning models theoret-
209 ically have advantages in pattern recognition.

210 7. Limitations and Future Work

211 7.1. Limitations

212 Several limitations are recognized in this study. The data's scope is restricted to the Delhi region, and
213 its generalizability to other Indian cities requires confirmation. There are limited meteorological variables
214 involved in weather data, and more features could increase accuracy. Holidays, special events, and eco-
215 nomic indicators are examples of external factors that were not taken into account. Batch forecasting is
216 the main focus of the study; real-time streaming adaptations have to be developed.

217 7.2. Future Research Directions

218 Future research should extend multi-region modeling to other Indian cities, investigate ensemble ap-
219 proaches that combine the best-performing models to increase accuracy, set up real-time learning for dy-
220 namic model updates, integrate renewable energy sources, including solar and wind generation forecasts,
221 and enhance the modeling capabilities for extreme events to handle anomalous consumption patterns.

8. Conclusion

While traditional statistical methods offer baseline forecasting capabilities, this in-depth study illustrates that advanced machine learning approaches, namely Gradient Boosting with engineered lag features, outperform significantly the conventional techniques for forecasting Delhi's electricity load.

The main contributions are methodological innovation through: a systematic comparison of statistical, machine learning, and deep learning approaches combined with substantial feature engineering; practical significance through the attainment of 6.2% MAPE, which provides actionable accuracy for energy planning and grid operations; proof that the rolling statistics and lag variables can improve the model performance by over 20%; and evidence-based recommendations for the stakeholders in Delhi's energy sector.

The results show that ensemble approaches can outperform more complex deep learning architectures for this application in terms of forecasting performance when duly adjusted with domain-specific features. The present research has provided a sound foundation for implementing data-driven electricity forecasting systems in Delhi and other Indian cities with similar urban settings.

Acknowledgment

The authors used a synthetic data set, which was generated using Kaggle and based on publicly accessible trends from the Government of India Holiday Calendar, OpenWeather API, and the National Load Dispatch Centre (NLDC). Data were strictly used for research purposes only and shows realistic hourly power consumption patterns for the Delhi area. The open-access weather resources that facilitated this study are maintained by the India Meteorological Department (IMD), for which the authors are also thankful.

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